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NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #1

Introduction to Chart of Nuclides, Nuclear Reactions and Radioactive Decay Modes

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NE235 Nuclear Reactor Operations Training

Module #1

Course Introduction – Review Syllabus

Module #1 Contents:

1. Introduction to Chart of Nuclides
2. Radioactive Decay Modes
3. Nuclear Reactions involving Neutrons
 - Neutron Energy Spectrum
 - Energy Dependent Neutron Cross Sections
 - Example Reactions

Laboratory Session this week:

- Introduction to PULSTAR Reactor Systems

Homework #1&2 – Due 9/6/2022

NE235 Nuclear Reactor Operations Training

Module #1

Fundamental basis of all nuclear engineering calculations is the
Reaction Rate Equation:

$$\text{Reaction Rate (1/sec)} = N \times \sigma \times \phi$$

where:

N = number of atoms (of a particular isotope) (# atoms)

ϕ = neutron flux in neutrons per cm² per second (n/cm²/sec)

σ = cross section; probability that reaction will occur in barns (b) (where 1 b = 1x10⁻²⁴ cm²)

SO, in Nuclear Engineering we need to understand:

- What nuclei are neutrons reacting with?
- How are they reacting and with what probability?
- What are the products of these reactions and what forms of matter and energy (e.g. radiation) do they produce?

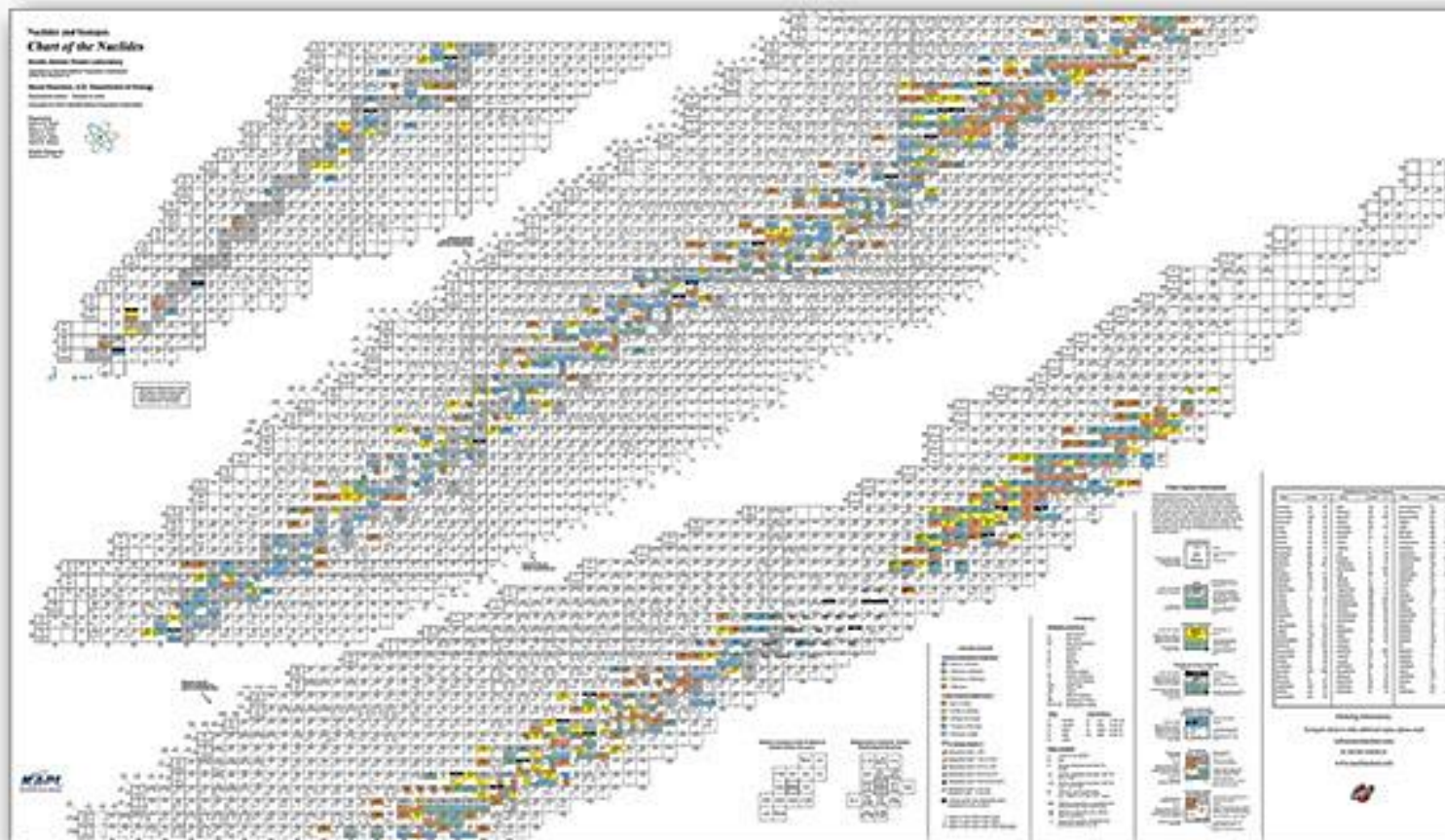
Chart of the Nuclides

Created at National Superconducting Cyclotron Laboratory, Michigan State University, 2013 using L^AT_EX 4.4

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Operation of NSC, as a national user facility, is supported by the Experimental Nuclear Physics Program of the U.S. National Science Foundation.

Chart of Nuclides



Knolls Atomic Version – avail. in chart or booklet form

Also online databases, e.g.:

<http://nucleardata.nuclear.lu.se/toi/index.asp>

Chart of Nuclides - KEY

Layout:

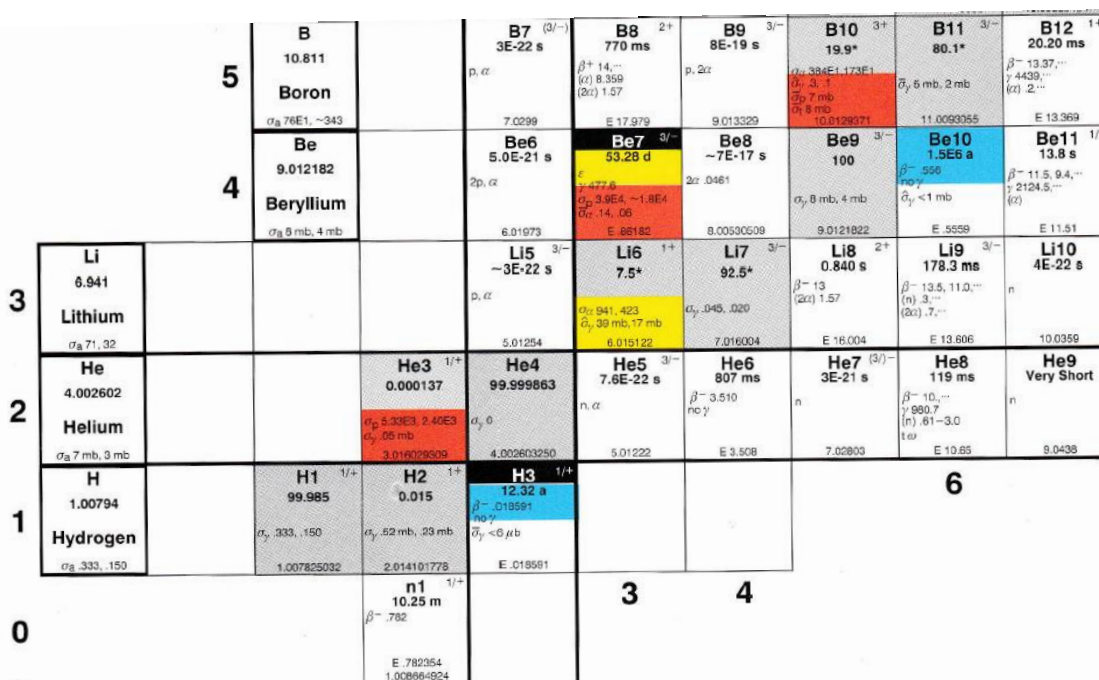
Z (y-axis) = atomic #;
number of protons

N (x-axis) = number of
neutrons

Isotopes in rows (have
same # of protons)

Isotones in columns (have
same # of neutrons)

Atomic Mass Number
 $A = Z + N$



Z ↑

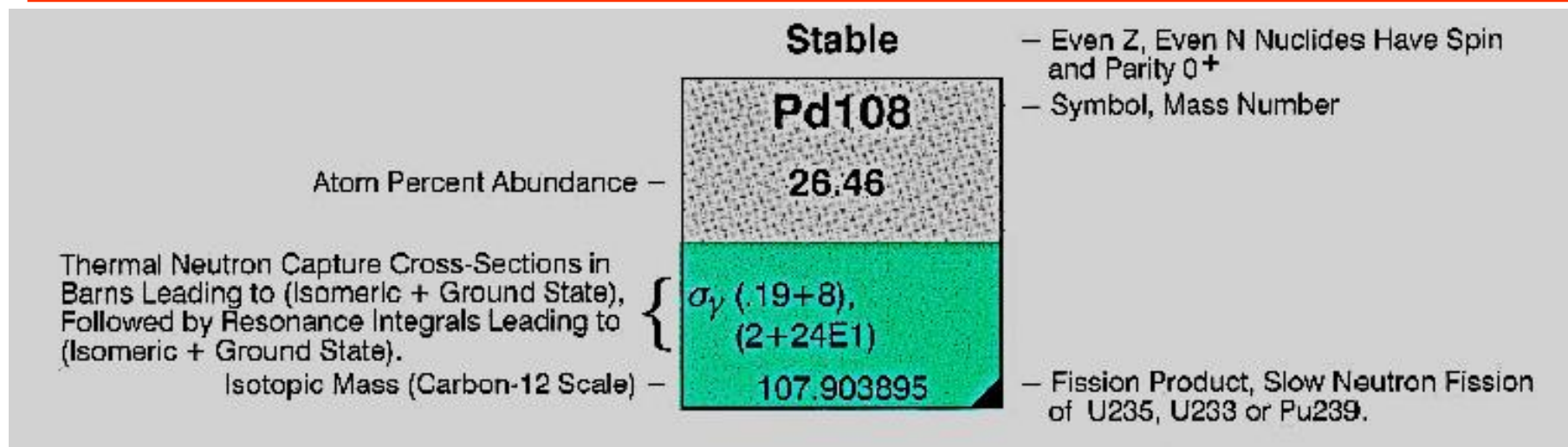
→ **N**

Chart of Nuclides - KEY

Element Block: At left end of each row

Chemical Element	
H 1.00794 Hydrogen σ_a .333, .150	– Symbol – Atomic Weight (Carbon-12 Scale) – Element Name – Thermal Neutron Absorption Cross-Section in Barns Followed by Resonance Integral, in Barns

Chart of Nuclides - KEY



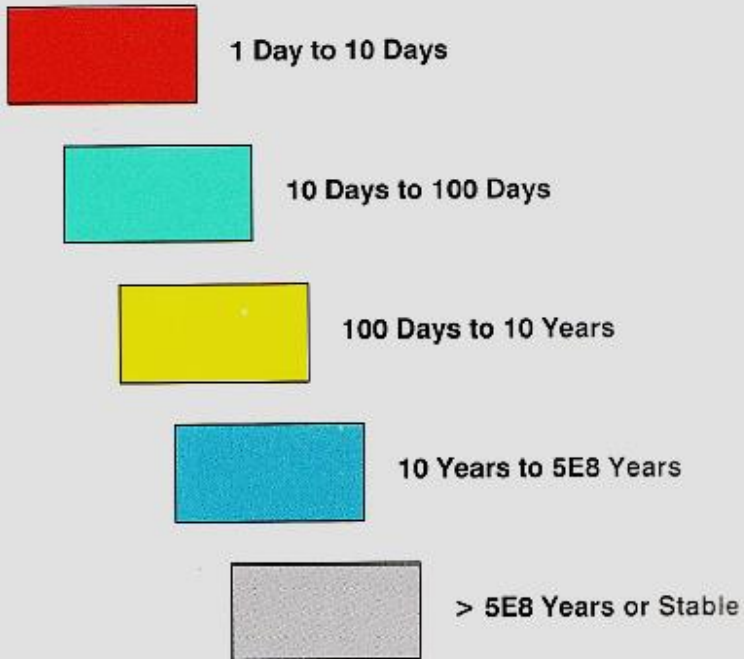
Stable Isotopes:

- Grey block on top half; color of bottom half indicates magnitude of neutron σ .
- Elemental Symbol and Atomic Mass Number; $A = N+Z$ for the isotope
- Atom Percent Abundance (given for abundances in nature)
- Neutron Cross Sections in barns (thermal and resonance integrals), leading to isomeric and ground states (see Cobalt-60 slide for good example).
- If it is a potential fission product, it is noted by presence of **black triangle** in lower right corner.

Chart of Nuclides - KEY

Block Colors:

Colors Used for Half Lives (Appear in Upper Half of Nuclide Block)



Colors Used for Neutron Absorption Properties (Appear in Lower Half of Nuclide Block)

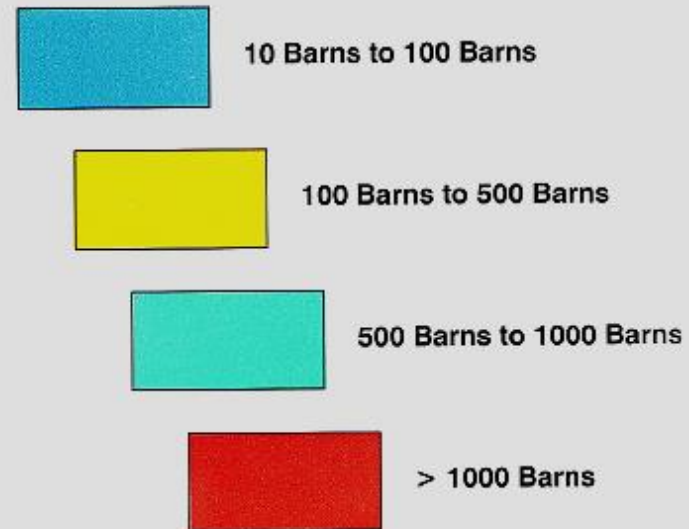
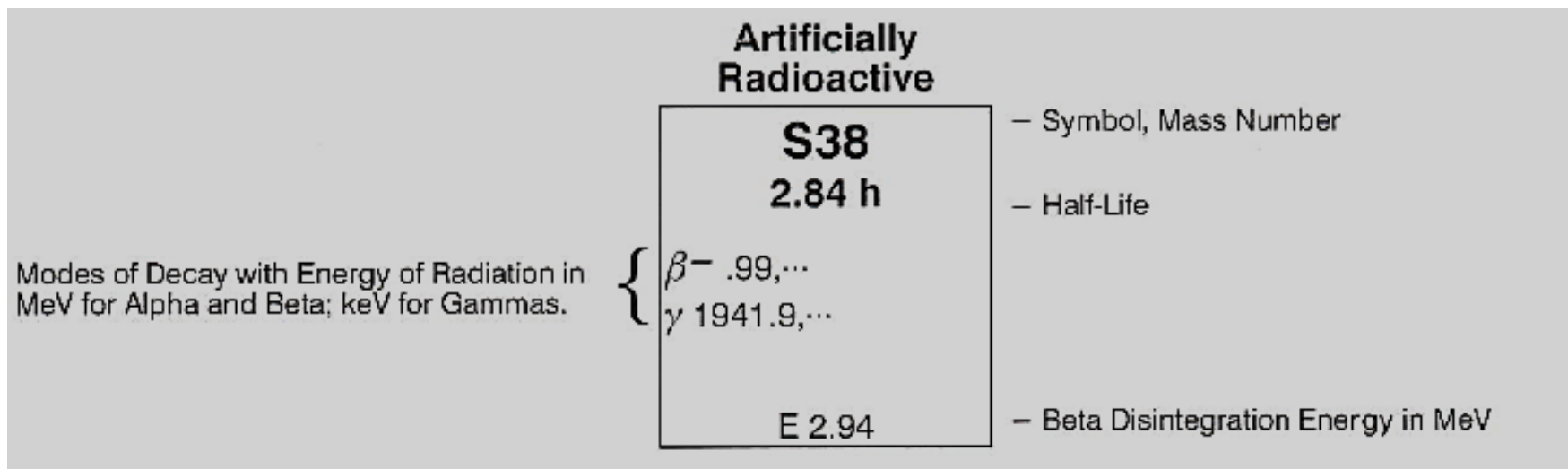


Chart of Nuclides - KEY



Artificially Radioactive Isotopes:

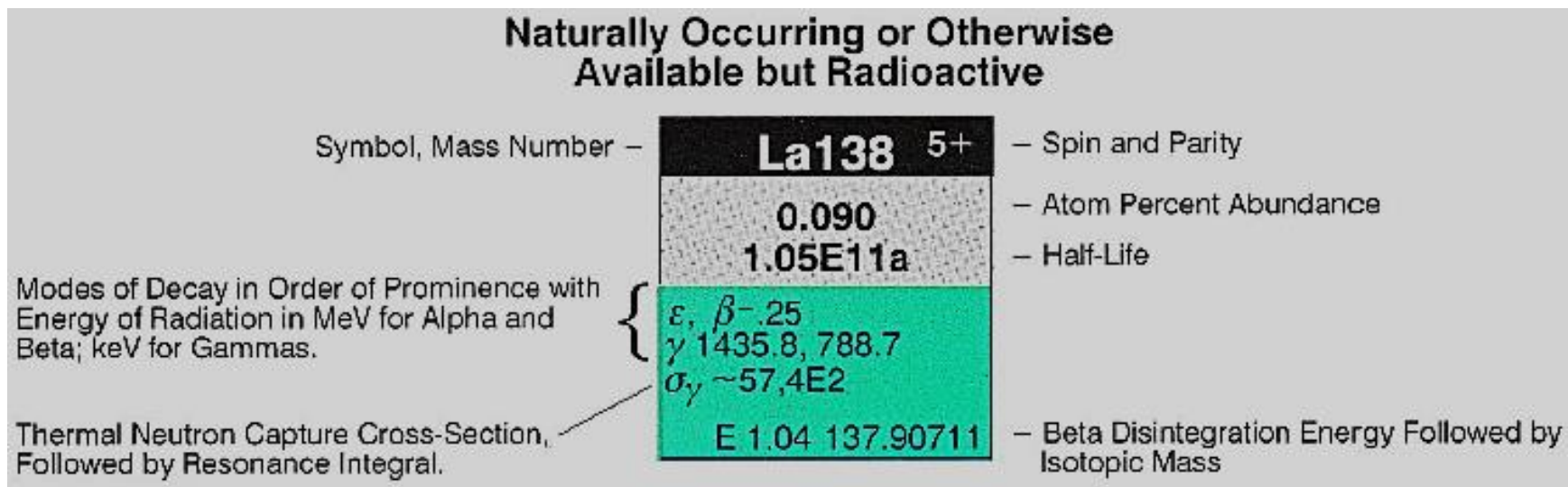
- White or colored block on top half, with color indicating magnitude of half life.
- Elemental Symbol and Atomic Mass Number $A = N + Z$ for the isotope
- Half life (sec., min., hours, days, or years (a = annum))
- Decay modes (α , β , γ , n, EC, IT, etc...) + energy of radioactive emission.

Chart of Nuclides - KEY

Decay modes:

α	alpha particle
β^-	negative electron (beta decay)
β^+	positron
γ	gamma ray
n	neutron
p	proton
d	deuteron
t	triton
ϵ	electron capture
IT	isomeric transition
D	delayed radiation (see page 6)
SF	spontaneous fission
E	disintegration energy
e^-	conversion electron

Chart of Nuclides - KEY



Naturally Occurring Radioactive Isotopes:

- e.g. ^3H (Tritium)(12.3a), ^{14}C (5730a), ^{40}K (1.28E9a)
- Uranium-238 (4.47E9a) + Decay Series
- Thorium-232 (1.40E10a) + Decay Series
- Radium-226 (1.60E3a) + Decay Series $\rightarrow$ Radon-222 (3.82d)

Uranium Decay Series

[illegible]

U238

UI 99.2745
4.47E9 a

α 4.197, 4.147, ...
 γ 49.6 ω (e^-), ...
SF $\nu\omega$
 σ_γ 2.68, 277
 $\sigma_{\text{tot}}^{\text{p-p}}$ $\sim 5 \mu\text{b}$, 1.3 mb
 σ_{a} 1 μb
238.050785

Uranium Decay Series

Uranium Series ($4n + 2$)*

Nuclide	Historical name	Half-life	Major radiation energies (MeV) and intensities†		
			α	β	γ
$^{238}_{92}\text{U}$ ↓	Uranium I	$4.51 \times 10^9 \text{ y.}$	4.15 (25%) 4.20 (75%)	---	---
$^{234}_{90}\text{Th}$ ↓	Uranium X ₁	24.1d	---	0.103 (21%) 0.193 (79%)	0.063c‡ (3.5%) 0.093c (4%)
$^{234}_{91}\text{Pa}^m$ ↓	Uranium X ₂	1.17m	---	2.29 (98%)	0.765 (0.30%) 1.001 (0.60%)
99.87% ↓ 0.13% ↓ $^{234}_{91}\text{Pa}$	Uranium Z	6.75h	---	0.53 (66%) 1.13 (13%)	0.100 (50%) 0.70 (24%) 0.90 (70%)
$^{234}_{92}\text{U}$ ↓	Uranium II	$2.47 \times 10^5 \text{ y}$	4.72 (28%) 4.77 (72%)	---	0.053 (0.2%)
$^{230}_{90}\text{Th}$ ↓	Ionium	$8.0 \times 10^4 \text{ y}$	4.62 (24%) 4.68 (76%)	---	0.068 (0.6%) 0.142 (0.07%)
$^{226}_{88}\text{Ra}$	Radium	1602y	4.60 (6%) 4.78 (95%)	---	0.186 (4%)

Neutron Energy Spectrum and energy dependent reaction Cross Section

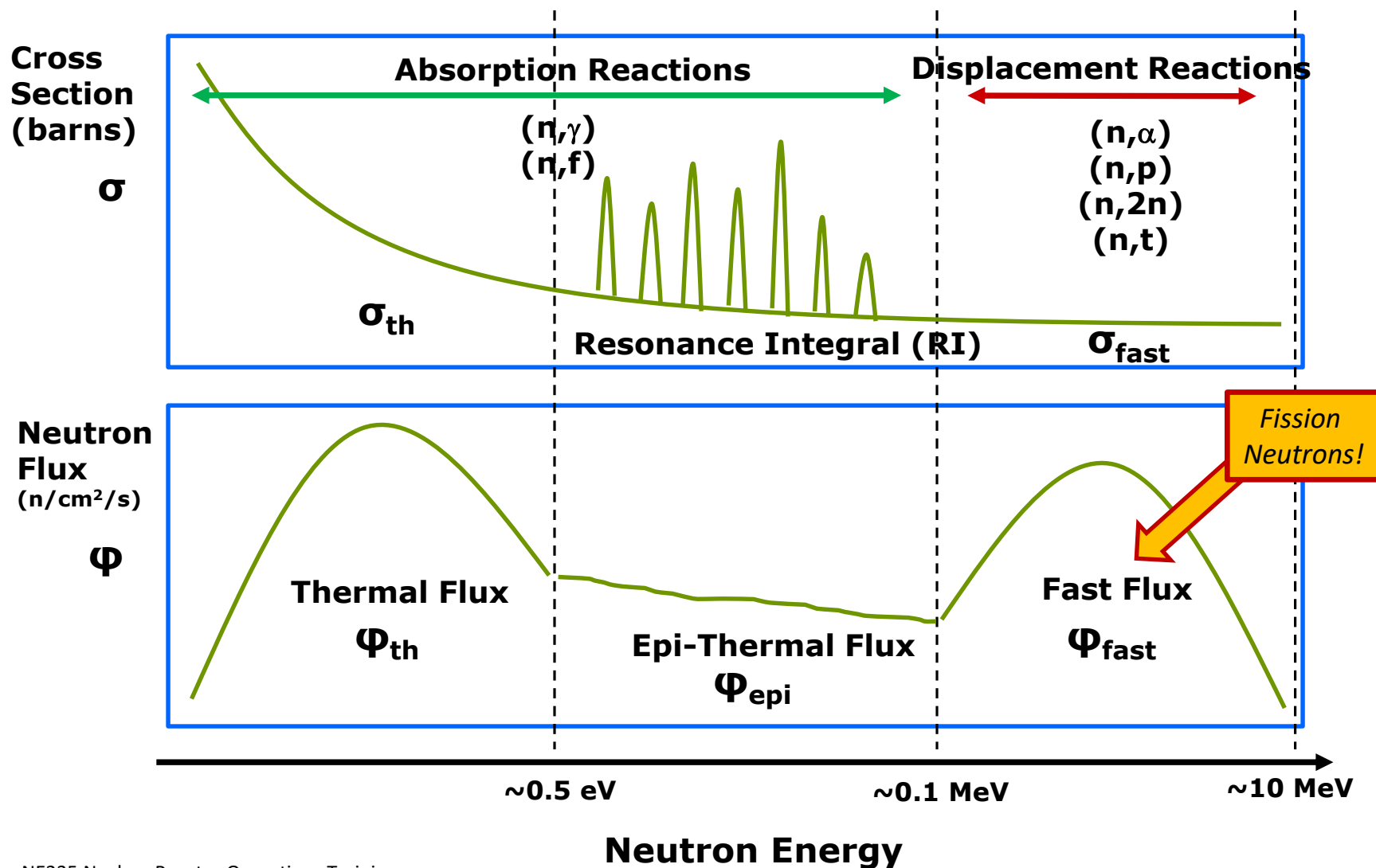


Chart of Nuclides - KEY

Displacement Caused by Nuclear Bombardment Reactions

	$\alpha, 3n$	$\alpha, 2n$ ${}^3\text{He}, n$	α, n	
	p, n	p, γ d, n ${}^3\text{He}, np$	α, np t, n ${}^3\text{He}, p$	
	p, pn γ, n $n, 2n$	Original Nucleus n, n	d, p n, γ t, np	t, p
p, α	n, t γ, np n, nd	n, d γ, p n, np	n, p $t, {}^3\text{He}$	
	n, α $n, n {}^3\text{He}$	$n, {}^3\text{He}$ n, pd		

Examples of Absorption Reactions:

- ${}^{59}\text{Co} (n, \gamma) {}^{60}\text{Co}$ (capture)
- ${}^{27}\text{Al} (n, \gamma) {}^{28}\text{Al}$ (capture)
- ${}^{238}\text{U} (n, \gamma) {}^{239}\text{U} \rightarrow {}^{239}\text{Np} \rightarrow {}^{239}\text{Pu}$ (capture/breeding)
- ${}^{235}\text{U} (n, \gamma) {}^{236*}\text{U} \rightarrow \text{FISSION!}$

Examples of Displacement Reactions:

- ${}^{27}\text{Al} (n, \alpha) {}^{24}\text{Na}$
- ${}^{54}\text{Fe} (n, p) {}^{54}\text{Mn}$
- ${}^{10}\text{B} (n, p) {}^{10}\text{Be}$
- ${}^{10}\text{B} (n, \alpha) {}^7\text{Li}$
- ${}^{10}\text{B} (n, t) {}^8\text{Be}$

Example: $^{59}\text{Co} (n,\gamma) ^{60}\text{Co}$ Capture Reaction

Stable

Radioactive ^{60}Co & ^{60m}Co

Co59 7/-			
100			
σ_γ	(21-16)	(39-35)	
58.933200			

(n,γ) capture to ground state

(n,γ) capture to isomeric state

R.I. to ground state

R.I. to isomeric state

σ_{thermal} to ground state

σ_{thermal} to isomeric state

2+	Co60	5+
10.47 m		5.271 a
IT 58.6, e ⁻		β^- .318,
β^- 1.6 ω, ...		...
γ 1332.5		γ 1332.5,
ω, ...		1173.2,
...		...
σ_γ SE1,		σ_γ 2.0, 4
2.3E2		E 2.824

Isomeric State:
 ^{60m}Co

Ground State:
 ^{60}Co

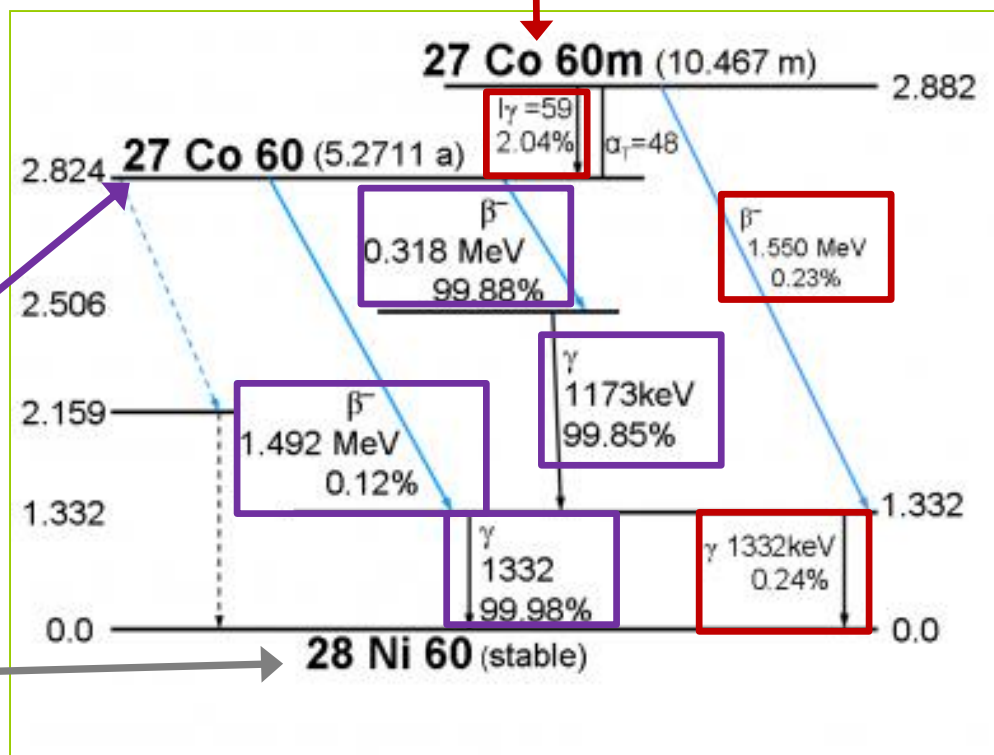
Example: ^{60}Co & $^{60\text{m}}\text{Co}$ Decay Modes

2+	Co60	5+
10.47 m		5.271 a
IT 58.6, e ⁻		β^- .318,
β^- 1.6 ω , ¹		γ 1332.5,
γ 1332.5		1173.2,
ω , ²		...
σ_γ 6E1,		σ_γ 2.0, 4
2.3E2		E 2.824

Isomeric State: $^{60\text{m}}\text{Co}$
 β^- , γ , and I.T. decay modes

Ground State: ^{60}Co
 β^- , γ decay modes

^{60}Ni Stable State
 (not radioactive)



Example: $^{238}\text{U} \rightarrow ^{239}\text{Pu}$ Breeding Reaction

Pu237 $7/2^-$ 45.2 d α 59.5, ... α 5.344 (e ⁻), ... γ 280.4 (v ω), 289.9, 320.8, ... σ_f 24E2 E_{α} .22	Pu238 87.7 a α 5.4992, 5.4565, ... γ 43.5 (e ⁻), ... γ 99.9 (e ⁻), ... SF v ω σ_{γ} 54E1, 20E1 σ_f 18, -33 σ_{α} <0.4 mb σ_{α} 238.049553	Pu239 $1/2^+$ 2.410E4 a α 5.156, 5.144, 5.105, ... γ 51.6 e ⁻ , 30.1-1057.3 (e ⁻), ... SF v ω σ_{γ} 271, 20E1 σ_f 750, 30E1 σ_{α} <0.4 mb σ_{α} 239.052156	Pu240 6.56E3 a α 5.1683, 5.1237, ... γ 45.2 (e ⁻), ... γ 104.2 (e ⁻), ... β^- .31E2 σ_{α} 2.4 σ_{α} .053808	Pu241 $5/2^+$ 14.4 a β^- .0206 α 4.897 (e ⁻), 4.853, ... γ 148.8 (v ω), 103.7, ... σ_{γ} -361, 16E1 σ_f 101E1, 57E1 σ_{α} <0.2 mb σ_{α} E .021	Pu242 3.75E5 a α 4.901, 4.856, ... γ 44.9 (e ⁻), ... SF v ω σ_{γ} 19, 11E2 σ_f <0.2, 0.23 σ_{α} 242.058737
Np236 (6^-) 22.5 h 1.55E5 a ϵ , β^- 54, ... γ 642.3, 687.6, ... σ_f 2.7E3, 7E2 E_{α} .914 E_{α} -48	Np237 $5/2^+$ 2.14E6 a α 4.788, 4.771, ... γ 29.4, 66.5, ... σ_{γ} 15E1, 65E1 σ_f 0.02, 7 σ_{α} 237.048166	Np238 2^+ 2.117 d β^- .263, 1.248, ... γ 984.5, 1028.5, ... σ_f 21E2, 9E2 E 1.292	Np239 $5/2^+$ 2.355 d β^- .438, 341, ... γ 106.1, 277.6, 228.2, ... σ_{γ} (3E1+3E1) σ_f <1 E .722	Np240 (5^+) 7.22 m 1.032 h β^- 2.18, β^- .89 γ 566.3, 973.9, 600.6 E 2.20	Np241 $5/2^+$ 13.9 m β^- 1.3, ... γ 174.9, 133.0 (e ⁻), 518.8, ... E 1.3
U235 $7/2^-$ 26 m AcU $IT \sim 76.8$ ev (v ω), e ⁻ α 4.395, 4.365, ... γ 185.7, 143.8, ... SF v ω σ_{γ} 99, 14E1 σ_f 585, -275 σ_{α} <.1 mb σ_{α} 235.043922	U236 2.342E7 a α 4.494, 4.445, ... γ 49.4 (e ⁻), 112.8, ... SF v ω σ_{γ} 5.1, 36E1 σ_f 0.04, 4 σ_{α} 236.045561	U237 $1/2^+$ 6.75 d β^- .24, .25, ... γ 59.5, 208.0, ... σ_{γ} 4E2, 12E2 σ_f <0.35 E .519	U238 99.2745 a α 4.197, 4.147, ... γ 49.6 (e ⁻), ... SF v ω σ_{γ} 2.68, 277 σ_f ~5 μ b, 1.3 mb σ_{α} 1 μ b σ_{α} 238.050785	U239 $5/2^+$ 23.47 m β^- 1.21, 1.28, ... γ 47.2, 2.5, ... σ_{γ}	U240 14.1 h β^- .36 γ 44.1 (e ⁻), ... E .39

U238 99.2745 a 4.47E9 a α 4.197, 4.147, ... γ 49.6 (e ⁻), ... SF v ω σ_{γ} 2.68, 277 σ_f ~5 μ b, 1.3 mb σ_{α} 1 μ b σ_{α} 238.050785
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238U:
 Fissionable
 Fertile
 NOT Fissile!

U235 $7/2^-$ 26 m AcU $IT \sim 76.8$ ev (v ω), e ⁻ α 4.395, 4.365, ... γ 185.7, 143.8, ... SF v ω σ_{γ} 99, 14E1 σ_f 585, ~275 σ_{α} <.1 mb σ_{α} 235.043922

235U & 239Pu
 Fissile

Pu239 $1/2^+$ 2.410E4 a α 5.156, 5.144, 5.105, ... γ 51.6 e ⁻ , 30.1-1057.3 (e ⁻), ... SF v ω σ_{γ} 271, 20E1 σ_f 750, 30E1 σ_{α} <0.4 mb σ_{α} 239.052156

Summary & Assignment

- *We now have the tools to calculate Reaction Rates for nuclear reactions (e.g. capture and fission)!*
- *The Chart of Nuclides indicates properties of isotopes and what reactions they undergo with neutron irradiation.*
- *For radioactive nuclides, decay modes indicate reaction products.*

Laboratory Session this week:

- Introduction to PULSTAR Reactor Systems

Homework #1 – Due 9/6/2022

1/+	U235	7/-
26 m	AcU	
IT ~76.8ev (ω), e ⁻	0.720 7.04E8 a	
α 4.395, 4.365, ...		
γ 185.7, 143.8, ...		
SF $\nu\omega$	σ_{γ} 99, 14E1	
σ_f 585, ~275	γ	
σ_{α} <.1 mb		
235.043922		